Mind the Gap:

Spatial Analysis of Metro-Bus Integration and Connectivity in Pune

INTERDISCIPLINARY MAJOR PROJECT 2026
GROUP 38

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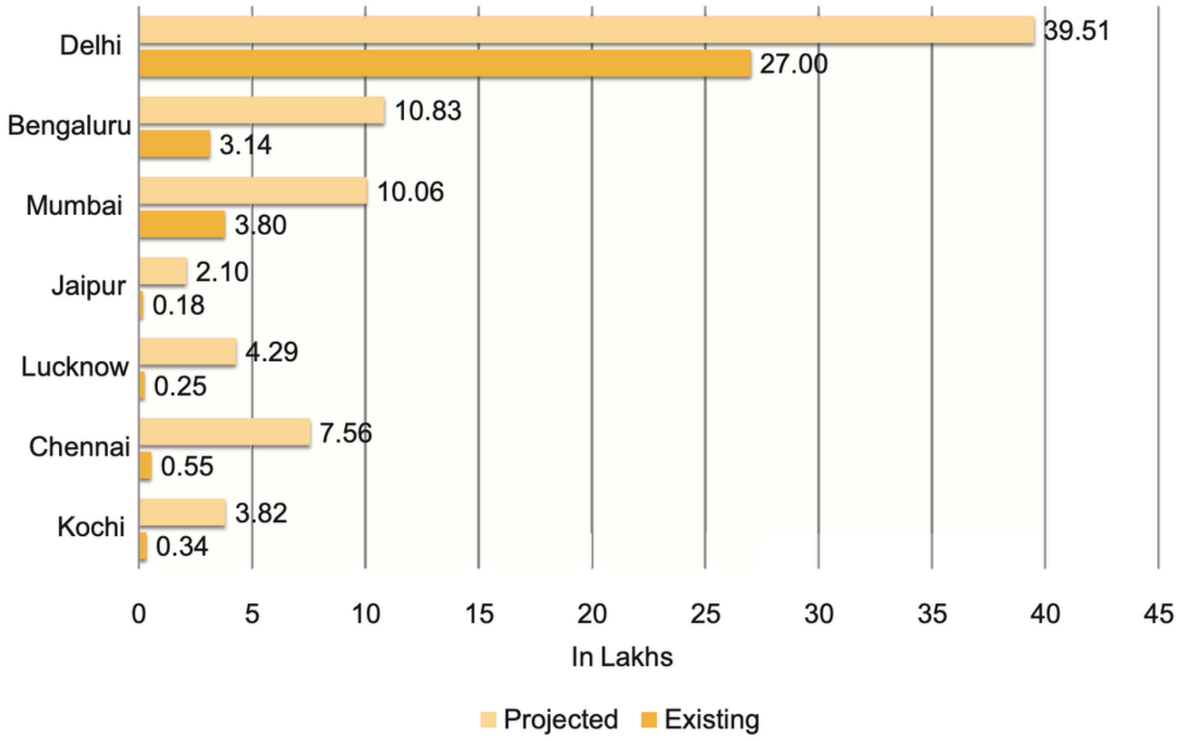
Figure 3:
Projected vs. Existing Ridership of Metro Rail Systems in India



Note: Projected ridership is subject to network completion. Ridership figures vary from 2014 to 2018.
Source: The Cost of Urban Commute: Balancing Affordability and Sustainability[13]

Public transportation systems account for 30 percent of trips in cities with populations between one and two million, 42 percent in areas with populations between two and five million, and 63 percent in cities with populations over five million (Irani, 2022).

Metro ridership across the country is far below the capacity, and millions are dependent on private vehicles to commute (Mukherjee et al, 2023; Irani, 2022)

Introduction

This is because

01 Integration across metro rail, city buses, suburban rail and feeder services are severely lacking

02 Reliable, safe, and affordable options for last mile connectivity are also sparse

While the number of cities in India increased threefold (from 2,363 to 7,935) and the urban population increased five times (from 79 million to 377 million) between 1961 and 2011, the number of private vehicles increased 200 times (from 0.7 million to 142 million) (Irani, 2022).
High usage of private vehicles have resulted in increasing travel time, traffic congestion, parking issues, road accidents, and risk to pedestrians.

We have observed a need for improvement within the public transportation system, especially route and network design, route allocation, coordination between different transport infrastructure, and data collection and management (Deb, 2002).

Literature Review

An efficient public transport system is said to: improve accessibility and economic opportunities (OECD, 2020); and reduce private vehicle dependence, congestion and carbon emissions (Mahadevia et al., 2023). Transport planning should consider long-term travel and land-use patterns (Suzuki et al., 1998).

WHY AN INTEGRATED PUBLIC TRANSPORT SYSTEM MATTERS

While Metro offers high capacity and faster commute, but limited coverage and high capital costs (Wei et al., 2020), Buses offer a wider geographic coverage. Hence, integrating these complementary modes can maximise commuter efficiency (Kumar et al., 2013).

INDIAN CONTEXT

01 Rapid urbanisation and severe traffic congestion create an urgent need for efficient public transport (TomTom Traffic Index, 2025; Dawda et al., 2021)

02 First- and last-mile connectivity is crucial for successful public transport (Srinivasa & Murthy, 2024)

03 Feeder and shared mobility can strengthen multimodal integration (Mukherjee et al., 2023; Srinivasa & Murthy, 2024)

Literature Review

PUNE CONTEXT

01

Metro, PMPML buses, BRT and rail often operate in isolation, creating service overlaps and service gaps (Sonar et al., 2023).

02

Some policy recommendations include multimodal integration, feeder services and integrated fares (PMRDA, 2024) to curb this issue.

LITERATURE GAP

Limited research evaluates whether Pune's current public transport network is efficient or compares metro investment with alternatives such as bus services.

Objectives

Primary

Study the complementarity between metro and bus routes in Pune, such that public transportation can be well connected and more citizens are encouraged to use it over private vehicles.

Aim1

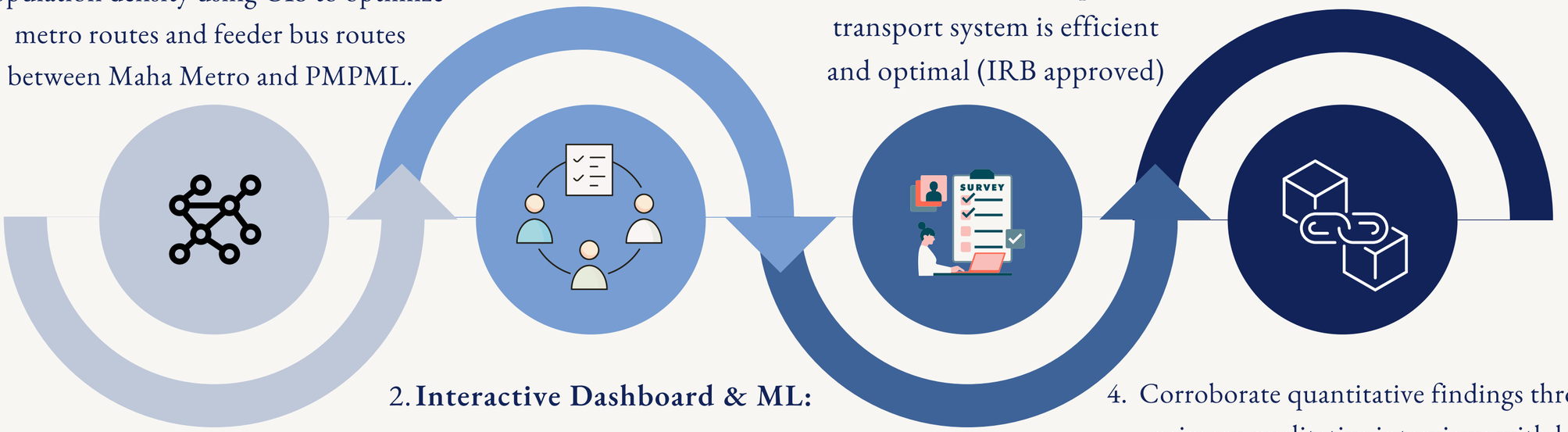
Design transportation infrastructure - specifically bus and metro - to align with the needs of the people and the spatial distribution of the city.

Aim 2

Simulate optimal routes and networks to provide potential feeder routes that would achieve a greater connectivity for citizens while optimising government resources.

Methodology

1. **Spatial & Network Analysis:**
Combines Open Streets Maps data and population density using GIS to optimize metro routes and feeder bus routes between Maha Metro and PMPML.

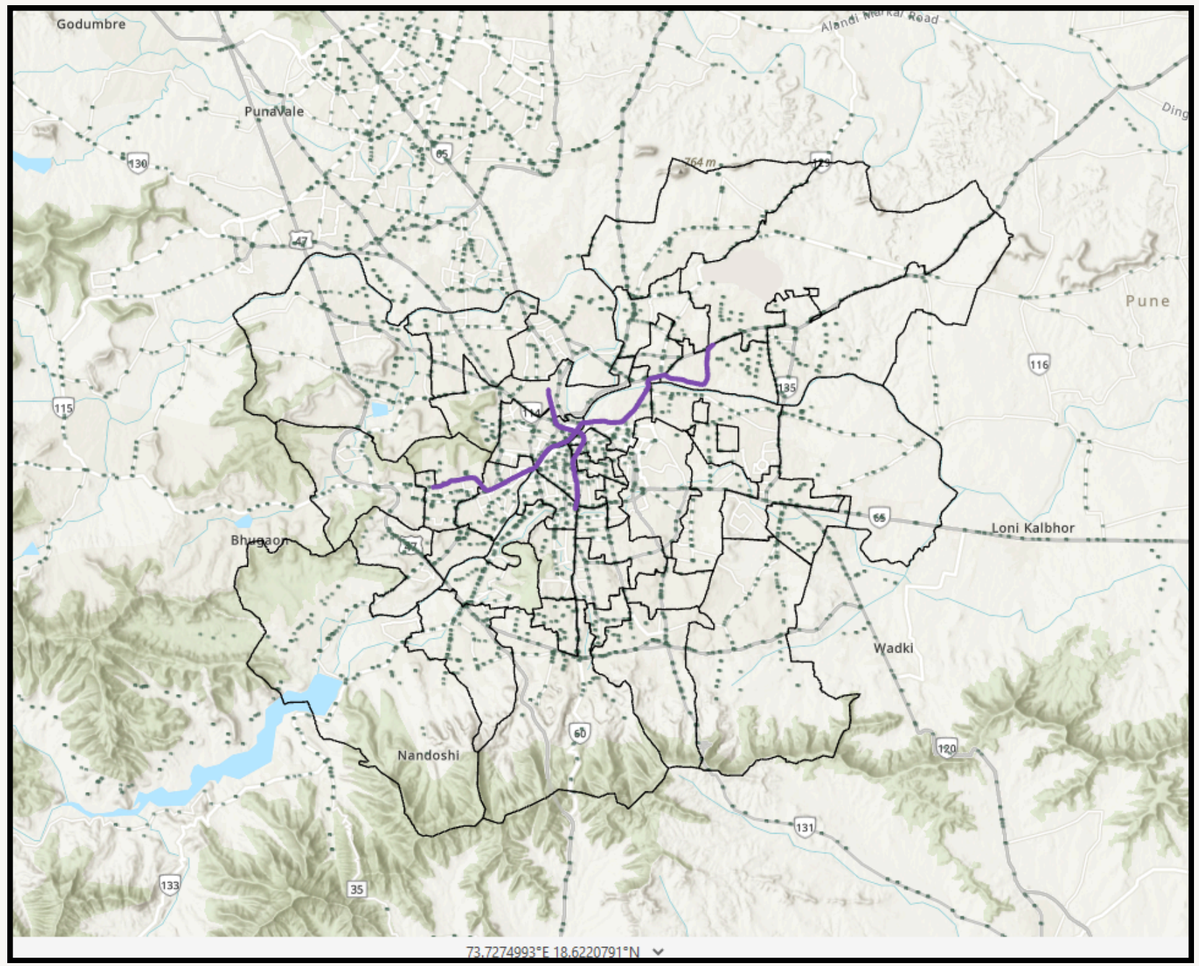


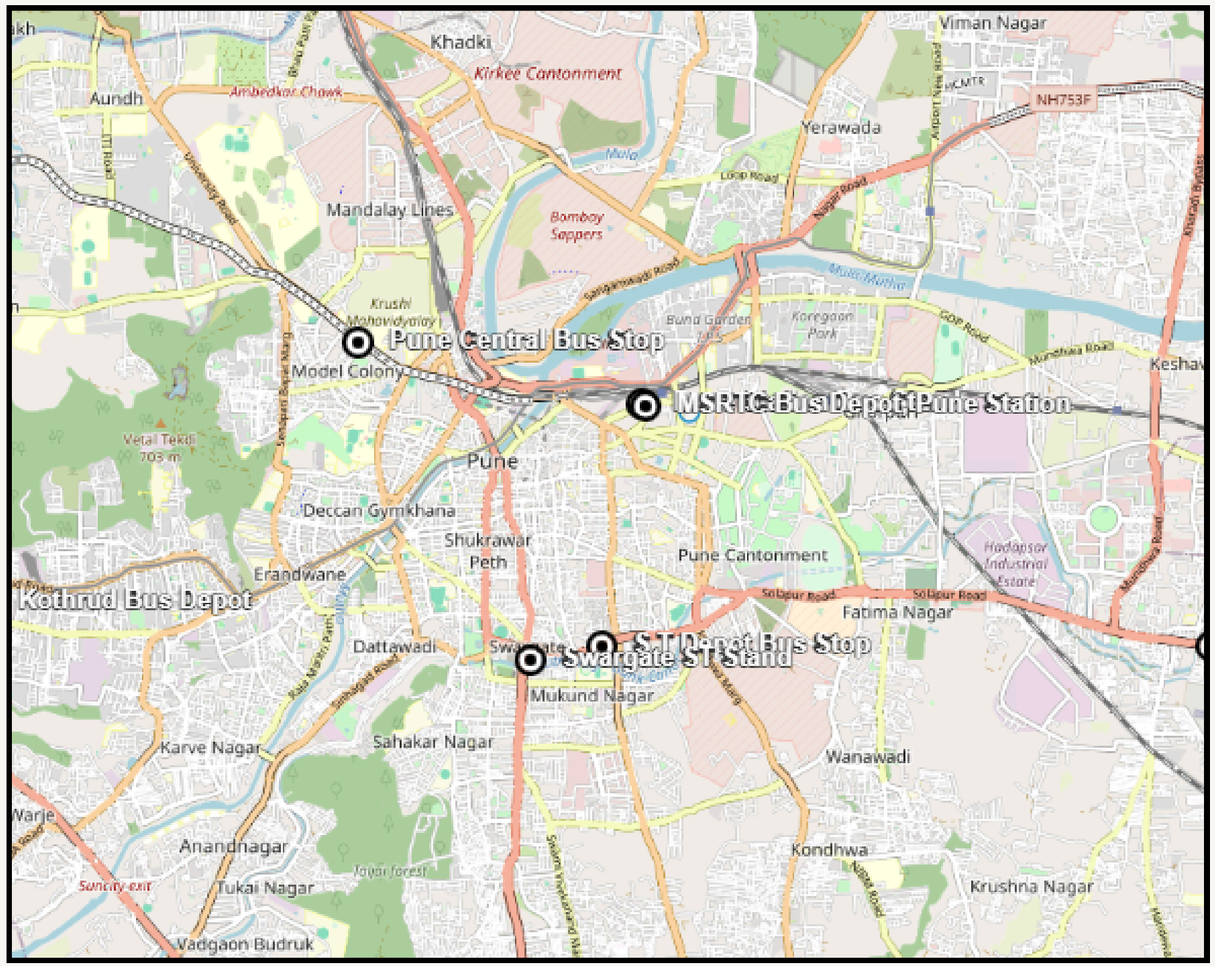
2. **Interactive Dashboard & ML:**
Integrates spatial overlays with machine learning models into a Plotly web application to monitor transfer delays across major transit hubs - Multi-Criteria Analysis

3. Survey of commuters in Pune to understand if the public transport system is efficient and optimal (IRB approved)

4. Corroborate quantitative findings through primary qualitative interviews with local transport officials (PMC, PMPML) and urban research organisations (WRI India, TERI) to ensure real world implementation feasibility.

Spatial Analysis





Survey Questions

Answer the questions that are relevant to you. The responses are anonymous, and completely voluntary.
उन सवालों के जवाब दें जो आपके लिए संगत हैं।जवाब गुमनाम और पूरी तरह से आपकी मर्ज़ी पर निर्भर हैं।

Section A

1. Age/उम्र
2. Primary Occupation/ व्यापार
3. Area of Residence/ निवास या घर
4. Area of Work/Study/ काम या पढ़ाई का क्षेत्र
5. Primary Mode of Transport (vehicle used for a majority of travel during the week)/ काम पर जाने के लिए परिवहन का साधन
6. Do you use more private transport (own vehicles, taxis, autorickshaws, cycle-rickshaws, etc) or do you use more public transport (metro, rail, bus, etc)? क्या आप ज़्यादातर टैक्सी, ऑटो-रिक्शा, साइकिल-रिक्शा, कार और स्कूटर वाहनों का इस्तेमाल करते हैं, या फिर मेट्रो, रेल और बस का?
7. Why?/ क्यों?

Section B

1. How often do you use the Pune Metro?/ आप पुणे मेट्रो का इस्तेमाल कितनी बार करते हैं?
2. Which days of the week do you use it normally? / आप हफ़्ते के किन दिनों में इसका सबसे ज़्यादा इस्तेमाल करते हैं?
3. What times of day do you use it most often? You can choose more than one. / आप दिन के किस समय इसका सबसे ज़्यादा इस्तेमाल करते हैं? आप एक से ज़्यादा विकल्प चुन सकते हैं।
4. What do you use it for?/ आप मेट्रो का इस्तेमाल किसलिए करते हैं?
5. Which metro station is the origin for your daily commute?/ आपके घर के सबसे पास कौन सा मेट्रो स्टेशन है?
6. Which metro station is closest to your destination for your daily commute?/ आपकी स्थान/ऑफ़िस के सबसे पास कौन सा मेट्रो स्टेशन है?
7. How do you reach these stations from your starting and ending points? (from home/from work)/ आप अपने घर या ऑफ़िस से इन स्टेशनों तक कैसे पहुँचते हैं?
8. How much time does it take to reach the nearest metro station?/ नज़दीकी मेट्रो स्टेशन तक पहुँचने में कितना समय लगता है?
9. How much time do you spend travelling in the metro?/ आप मेट्रो में यात्रा करने में कितना समय बिताते हैं?
10. How frequent are the metro cars on your route? Does it need to be more or less frequent?/ आपके रूट पर मेट्रो कितनी बार आती है? क्या यह कम या ज़्यादा होनी चाहिए?

Survey Questions

Section C

1. How often do you use PMPML or BRT buses?/ आप कितनी बार PMPML या BRT बसों का इस्तेमाल करते हैं?
2. Which days of the week do you use it normally?/ आप हफ़्ते के किन दिनों में इसका सबसे ज़्यादा इस्तेमाल करते हैं?
3. What time of the day do you use it most often? You can choose more than one./ आप दिन के किस समय इसका सबसे ज़्यादा इस्तेमाल करते हैं? आप एक से ज़्यादा विकल्प चुन सकते हैं।
4. What do you use it for?/ आप मेट्रो का इस्तेमाल किसलिए करते हैं?
5. Which bus stop is the closest to your home/point of origin? जहाँ से आप यात्रा शुरू कर रहे हैं (जैसे की आपका घर), वहाँ से सबसे पास कौन-सा बस स्टॉप है?
6. Which bus stop is the closest to your work/destination? / जहाँ से आप यात्रा अंत कर रहे हैं (जैसे की आपका कार्यालय), वहाँ से सबसे पास कौन-सा बस स्टॉप है?
7. How do you reach these bus stops from your home/work?/ आप अपने घर या ऑफ़िस से इन बस स्टॉप तक कैसे पहुँचते हैं?
8. How much time does it take you to reach these bus stops?/ नज़दीकी बस स्टॉप तक पहुँचने में कितना समय लगता है?
9. How much time do you spend travelling on the bus?/ आप बस में यात्रा करने में कितना समय बिताते हैं?
10. How frequent are the buses on your route? Does it need to be more or less frequent?/ आपके रूट पर बसों कितनी बार आती है? क्या यह कम या ज़्यादा होनी चाहिए?

Section D

1. Do you use multiple vehicles/modes of transport to commute?/ क्या आप एक दिन में आने-जाने के लिए एक से ज़्यादा गाड़ियों या ट्रांसपोर्ट का इस्तेमाल करते हैं?
2. Do you use both metro and bus while commuting?/ क्या आप आने-जाने के लिए मेट्रो और बस, दोनों का इस्तेमाल करते हैं?
3. If so, are the bus stops close to the metro stations? / क्या बस स्टॉप मेट्रो स्टेशनों के पास हैं?
4. If the bus stops are not close to the metro stations, what do you do instead? / अगर बस स्टॉप मेट्रो स्टेशनों के पास नहीं हैं, तो आप क्या करते हैं?
5. List areas/roads/names of stations or stops that have these issues of connectivity. / उन इलाकों, सड़कों या स्टेशनों/स्टॉप के नाम बताएं जो बस स्टॉप को मेट्रो स्टेशन से नहीं जोड़ते हैं।
6. What is the biggest challenge when switching between bus and metro while travelling through the city? / बस और मेट्रो के बीच बदलने में सबसे बड़ी कठिनाई क्या है?

Section E

1. Which areas/roads/stations/locations do you feel the most traffic congestion during your commute?/ आने-जाने के दौरान आपको किन इलाकों, सड़कों, स्टेशनों या जगहों पर सबसे ज़्यादा ट्रैफ़िक जाम का सामना करना पड़ता है?
2. Which is the main transport corridor where you think a new feeder bus route or metro connectivity would be the most useful for your daily commute? / आपके रोज़ाना आने-जाने के लिए किस मेट्रो स्टेशनों में नया फ़ीडर बस रूट सबसे ज़्यादा फ़ायदेमंद रहेगा?
3. Do you need to transfer between stations and if so, at which point (location pin) do you usually transfer between PMPML buses, or between Pune Metro stations, or between the two transport systems? / आप PMPML बसों, पुणे मेट्रो स्टेशनों या इन दोनों ट्रांसपोर्ट सिस्टम के बीच किस जगह पर बदलते हैं?
4. (Referring to 3) Does the main transit interchange station cause any inconvenience for shifting in terms of / स्टेशनों के बीच ट्रांसफर के दौरान, क्या इन मामलों में कोई परेशानी/असुविधा होती है:

(i) timing/frequency समय/आवृत्ति

(ii) crowd भीड़?

5. Select the exact spot or transfer station (in PMPML and Metro stations) where you experience the worst delay or inconvenience. / किस स्टेशन या बस स्टॉप पर आपको सबसे ज़्यादा देरी या परेशानी का सामना करना पड़ता है?
6. Is there only one transport route (buses or metros) operating as your route in your daily commute?/ क्या आपके रोज़ाना के सफ़र में सिर्फ़ एक ही ट्रांसपोर्ट रूट (बस या मेट्रो) चलता है?
7. Do you take a longer route or use private transport specifically to avoid a poorly connected transit route? If yes, specify locations (at least one). / क्या आप खराब कनेक्टिविटी वाले परिवहन से बचने के लिए कोई लंबा रास्ता अपनाते हैं या प्राइवेट ट्रांसपोर्ट का इस्तेमाल करते हैं? अगर हाँ, तो उन जगहों के नाम बताएं।

Interdisciplinary perspective

Six disciplines, six lenses on Pune’s public transportation

01 Computer Science + Business Analytics

- Analyse ridership trends, traffic congestion, and ward-wise population density
- Overlay existing PMPML bus routes with the proposed metro lines
- Ascertain the spatial coverage of Pune's public transport network

02 Economics + Finance

- Examine costs and explore funding: PPPs, government grants, fare revenue
- Evaluate whether recommendations are sustainable long-term
- Use transport-department revenue data to size how many extra buses can be deployed

03 Public Policy

- Study existing regulations and the roles of PMC and PMPML
- Develop realistic implementation plans grounded in governance
- Account for administrative processes and practical challenges

04 Sociology

- Study how different groups travel and the challenges they face daily
- Examine the city's living structure and spatial segregation in urban planning
- Ensure solutions are inclusive of low-income and non-tech-reliant communities

Proposed Solution

We propose to build an interactive, publicly accessible map overlaying Pune's metro and bus routes.

Since bus routes are more flexible than fixed metro lines, the tool helps identify overlap and gaps to guide new bus route planning

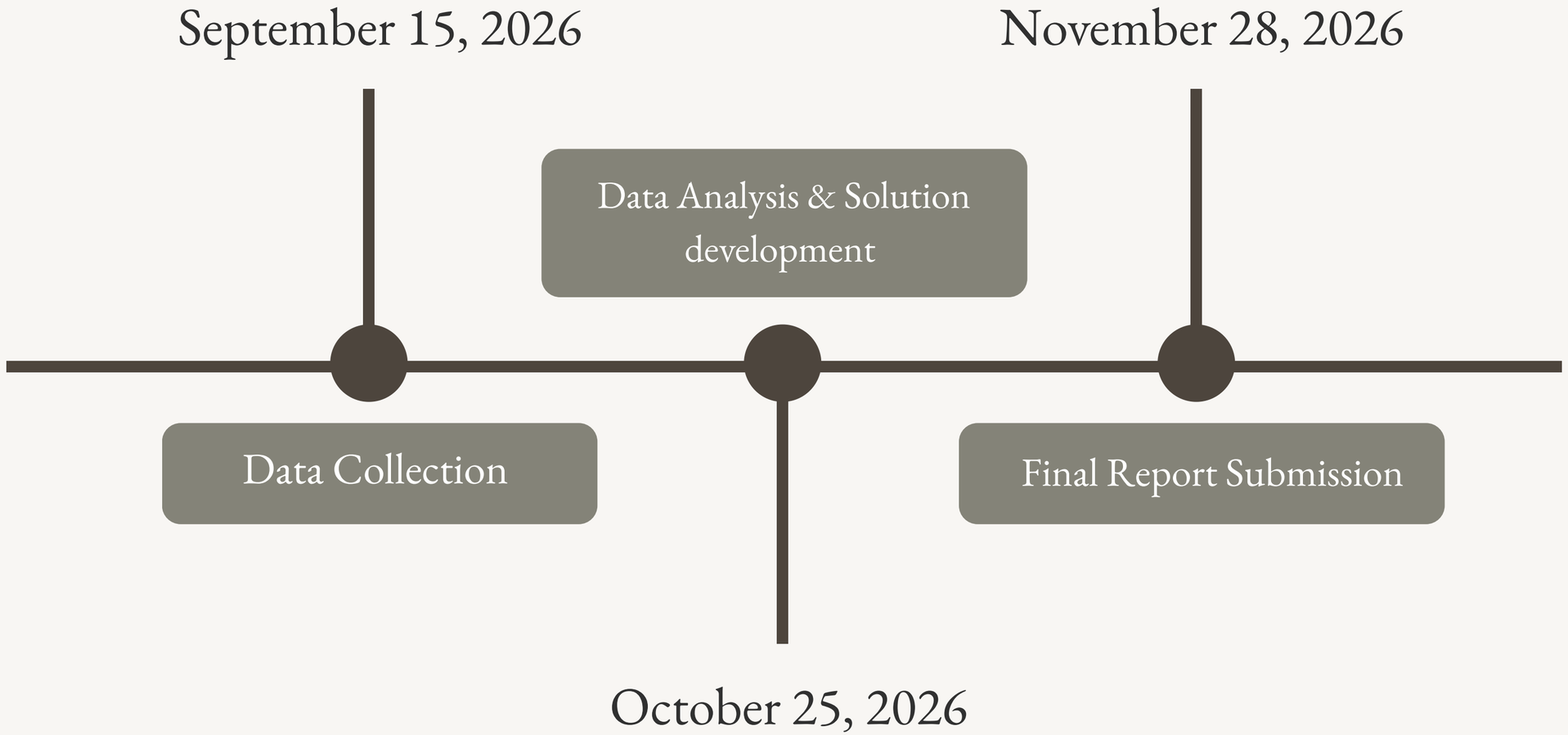
Dynamic map lets users plot custom routes, pulls live data via existing APIs, and factors in commuter destinations, wardwise population and volume by time of day.

Our primary user will be PMPML (Pune Mahanagar Parivahan Mahamandal Limited); stakeholders include PMC and the general public. We intend to identify uncovered localities, prioritize urgent feeder routes, flag redundant bus stops overlapping with metro and estimate infrastructure needs.

Low capital cost, relies on publicly available data, self-updating as new metro lines go live

Supports SDG 11 (Sustainable Cities), SDG 9 (Industry & Infrastructure), and indirectly SDG 13 (Climate Action) via reduced congestion and carbon footprint

Timeline


September 15, 2026
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Data Analysis & Solution development
Data Collection
Final Report Submission
October 25, 2026

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